System Overview

Job Recruitment and Internship Platform

1.1 Purpose

The Job Recruitment and Internship Platform is a web platform that connects job seekers and employers. It supports four main functions: job applications, CV management, candidate matching, and interview scheduling.

1.2 The Problem

Hiring today is slow and scattered:

- Applicants must search many websites and send their CVs again and again.
- Recruiters receive many CVs in different formats and must read them one by one.
- Applicants often get no confirmation or news about their application.
- CVs contain personal data, so they must be kept safe.

This platform puts the whole process in one place.

1.3 Who Uses It

User	What they do
Applicant	Searches for jobs and internships, uploads a CV, and applies
Recruiter	Publishes vacancies, reviews candidates, and schedules interviews
Platform Administrator	Checks job listings and manages users to keep the platform safe

1.4 External Systems

The platform depends on two outside services:

System	What it does for the platform
Email Service	Sends application confirmations and interview notifications
Apache Tika	Reads uploaded CV files and extracts the text, so the platform can match candidates to jobs

1.5 Why It Is Architecturally Interesting

The system looks simple, but it must solve several design challenges at the same time. Each one links to a quality attribute scenario in Section 5.

Challenge	Why it matters	Scenario
Fast search	Applicants search through thousands of vacancies and expect quick results	Performance (QAS-1)
Outside service failure	If the Email Service is down, applications must still be saved and emails sent later	Availability (QAS-2)
Data protection	CVs and candidate lists are private and must be safe from unauthorized access	Security (QAS-3)
Easy change	Apache Tika should be replaceable without changing the rest of the system	Modifiability (QAS-4)
Traffic peaks	Events like a university job fair can raise users from 500 to 5,000	Scalability (QAS-5)
Easy to use	First-time applicants must be able to apply quickly without help	Usability (QAS-6)

These challenges pull the design in different directions. For example, strong security checks can slow down search, and extra servers for peak traffic add cost. The architecture must balance them, and this makes the platform a good subject for architectural study.